

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Sixth Semester of B. Tech (EC) Examination****November- December 2015****EC310.01/EC310 Optical Communication****Date: 27.11.2015, Friday****Time: 01:30 p.m. To 04:30 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I**Q - 1 Answer the question below. [07]**

- a. Why fiber optic cable is more reliable than copper cable ? Give the reason in one sentence. [01]
- b. What is total internal reflection? [01]
- c. What is group delay ? [01]
- d. Define numerical aperture. [01]
- e. Give the difference between single mode and multi mode fiber. [01]
- f. What makes fiber optic immune to EMI? [01]
- g. List out three different mechanisms that causes absorption loss in fiber optics. [01]

Q – 2.a Calculate the numerical aperture of step index fiber having $n_1 = 1.48$ and $n_2 = 1.50$. [04]
 What is the maximum entrance angle $\theta_{0 \max}$ for this fiber, if the outer medium is air with $n = 1.00$?

Q – 2.b Answer any two questions. [10]

- 1) Discuss the key elements of optical fiber communication system.
- 2) What are the possible lensing schemes used to improve optical source to fiber coupling efficiency? Discuss with necessary diagram.
- 3) What do you mean by dispersion? Describe Chromatic dispersion in brief.

Q - 3 Answer the following.

- a. A double- heterojunction InGaAsP LED emitting at a peak wavelength of 1310 nm has radiative and nonradiative recombination times of 25 and 90ns , respectively. The drive current is 35 mA.($h = 6.625 \times 10^{-34}$ J*s , $c = 3 \times 10^8$ m/s , $q = 1.9 \times 10^{-19}$ C) [05]
 - 1) Find the internal quantum efficiency and internal power level.
 - 2) If the refractive index of the light source material is $n = 3.5$, find the power emitted from the device.

- b. 1) Explain the terms Quantum efficiency and LED power. [05]
 2) Briefly explain equilibrium NA. [04]

OR

- 1) Explain three key transition processes involved in LASER action. [05]
 2) Enumerate the mechanical misalignment problem in fiber optics. [04]

SECTION – II

Q - 4 Answer the question below. [07]

- a. Define OTDR. [01]
 b. DWDM stands for _____. [01]
 c. Define splitting ratio in case of optical coupler. [01]
 d. Give the importance of ADM. [01]
 e. What is switching? Give the Difference between Packet switching and Brust switching. [02]
 f. Define impact ionization. [01]

Q – 5.a Write a short note on photo detector noise. [05]

OR

Q – 5.a Explain PIN photo detector with structure, principle of operation and the performance. [05]
 Compare it with RAPD.

Q – 5.b A given silicon avalanche photodiode has a quantum efficiency of 65 % at a wavelength of 900 nm. Suppose 0.5 μ W of optical power produce a multiplied photo current of 10 μ A, then find the multiplication factor M. [04]

Q – 5.c Draw the schematic diagram of typical optical receiver and also explain different type of preamplifier in short. [05]

OR

Q – 5.c Discuss optical power loss model for point to point link with necessary diagram. [05]

Q – 6.a Which factors affect the Rise Time Budget? In One optical communication link have the photo diode which required receiver input signal of -42dbm and optical source that can transmit -13dbm average optical power level in to the fiber fly head with 50 μ m core diameter. Assume that 1db loss occur when the fiber fly head is connected to the cable and another 1db connector loss. Link also includes the 6db system margin. Find the possible transmission distant with $\alpha_f = 3.5\text{db/km}$ [04]

Q – 6.b Explain SONET frame structure Give the difference between SONET and SDH. **[05]**

Q – 6.c Define EDFA. What are the key system features of WDM? **[05]**

OR

Q – 6.a Define SONET rings and explain two fiber BLSR ring. **[05]**

Q – 6.b Describe Mach-Zehnder interferometer with required sketch and equations. **[05]**

Q – 6.c Explain Subcarrier multiplexing. **[04]**
